

ASSESSMENT I: DATA INTERPRETATION

Question 1: Interpret the XML Structure

a. Root element:

The root element of the XML document is `<courses>`. It contains all the course records.

b. Repeating record element:

The repeating record element is `<course>`. There are five course records in the XML document.

c. Attribute used to uniquely identify each course:

The id attribute uniquely identifies each course. Examples are C101, C102, C103, C104, and C105.

d. Elements representing numeric information:

The numeric elements are:

- `<students>` – represents the number of students enrolled.
- `<credits>` – represents the number of course credits.

e. Is the XML document structurally well-formed?

Yes, the XML document is structurally well-formed because:

1. It has only one root element, `<courses>`.
2. All elements have proper opening and closing tags.
3. Elements are properly nested.
4. Attribute values are enclosed in quotation marks.
5. The XML declaration is correctly specified.

Question 2: Apply XPath for Data Selection

a. All course records

`/courses/course`

b. Names of all courses

`/courses/course/name`

c. Courses having more than 50 students

`/courses/course[students > 50]`

d. Courses carrying 4 credits

/courses/course[credits = 4]

e. Courses whose type is Theory

/courses/course[type = 'Theory']

f. Names of Theory courses having more than 50 students

/courses/course[type = 'Theory' and students > 50]/name

g. Faculty members handling courses with at least 4 credits

/courses/course[credits >= 4]/faculty

h. The course whose id is C104

/courses/course[@id = 'C104']

i. The first course available in the XML document

/courses/course[1]

j. The last course available in the XML document

/courses/course[last()]

Question 3: Apply XSLT for Data Presentation

The following XSLT stylesheet displays courses having more than 40 students and sorts them in descending order of student enrollment.

```
<?xml version="1.0" encoding="UTF-8"?>

<xsl:stylesheet version="1.0"
xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:output method="html" indent="yes"/>

<xsl:template match="/">

<html>
<head>
<title>High Enrollment Courses</title>
</head>

<body>

<h1>High Enrollment Courses</h1>
```

```

<table border="1">

<tr>
<th>Course Code</th>
<th>Course Name</th>
<th>Faculty</th>
<th>Students</th>
<th>Credits</th>
<th>Type</th>
</tr>

<xsl:for-each select="courses/course[students &gt; 40]">

<xsl:sort select="students"
data-type="number"
order="descending"/>

<tr>

<td>
<xsl:value-of select="code"/>
</td>

<td>
<xsl:value-of select="name"/>
</td>

<td>
<xsl:value-of select="faculty"/>
</td>

<td>
<xsl:value-of select="students"/>
</td>

<td>
<xsl:value-of select="credits"/>
</td>

<td>
<xsl:value-of select="type"/>
</td>

</tr>

</xsl:for-each>

</table>

</body>
</html>

</xsl:template>

</xsl:stylesheet>

```

XPath condition used in XSLT

`courses/course[students > 40]`

This condition selects only courses having more than 40 students.

Sorting used in XSLT

```
<xsl:sort select="students"
  data-type="number"
  order="descending"/>
```

This sorts the courses from the highest number of students to the lowest number of students.

Expected output

Course Code	Course Name	Faculty	Students	Credits	Type
AI302	Artificial Intelligence	Dr. Meena	72	4	Theory
ML304	Machine Learning	Dr. Priya	64	4	Theory
WEB301	Web Technology	Dr. Arun	58	4	Theory
DB305	Database Systems	Dr. Kumar	42	3	Theory

The Web Technology Laboratory course is not displayed because it has only 36 students.

Question 4: Interpret the Extracted Data

a. Course with the highest enrollment

The course with the highest enrollment is **Artificial Intelligence (AI302)** with **72 students**.

b. Course with the lowest enrollment

The course with the lowest enrollment is **Web Technology Laboratory (WEB303)** with **36 students**.

c. Number of Theory courses

There are **4 Theory courses**:

1. Web Technology
2. Artificial Intelligence
3. Machine Learning
4. Database Systems

Therefore, the number of Theory courses is **4**.

d. Courses having exactly 4 credits

The courses having exactly 4 credits are:

1. **Web Technology (WEB301)**
2. **Artificial Intelligence (AI302)**
3. **Machine Learning (ML304)**

Therefore, there are **3 courses with 4 credits**.

e. Courses requiring additional teaching assistant support

An additional teaching assistant is assigned to every course with more than 60 students.

The courses satisfying this condition are:

1. **Artificial Intelligence (AI302)** – 72 students
2. **Machine Learning (ML304)** – 64 students

Therefore, **AI302 and ML304 require additional teaching assistant support**.